In [2]:
```
import pandas as pd
```

In [3]:
```
import numpy as np
```

In [4]:
```
from sklearn.datasets import load_iris

iris = load_iris()

df = pd.DataFrame(iris.data,columns=iris.feature_names)

df["species"] = pd.Categorical.from_codes(
    iris.target,
    iris.target_names
)

print(df)
```

```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
\
0                  5.1               3.5                1.4               0.2
1                  4.9               3.0                1.4               0.2
2                  4.7               3.2                1.3               0.2
3                  4.6               3.1                1.5               0.2
4                  5.0               3.6                1.4               0.2
..                 ...               ...                ...               ...
145                6.7               3.0                5.2               2.3
146                6.3               2.5                5.0               1.9
147                6.5               3.0                5.2               2.0
148                6.2               3.4                5.4               2.3
149                5.9               3.0                5.1               1.8

       species
0       setosa
1       setosa
2       setosa
3       setosa
4       setosa
..         ...
145  virginica
146  virginica
147  virginica
148  virginica
149  virginica

[150 rows x 5 columns]
```

In [5]:
```
print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

  species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

In [6]:
```
print("Last 5 rows:")
print(df.tail())
```

```
Last 5 rows:
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
\
145                6.7               3.0                5.2               2.3
146                6.3               2.5                5.0               1.9
147                6.5               3.0                5.2               2.0
148                6.2               3.4                5.4               2.3
149                5.9               3.0                5.1               1.8

       species
145  virginica
146  virginica
147  virginica
148  virginica
149  virginica
```

In [7]:
```
print("Dataset information:")
df.info()
```

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    category
dtypes: category(1), float64(4)
memory usage: 5.1 KB
```

In [8]:
```
print("Dataset shape:")
print(df.shape)
```

```
Dataset shape:
(150, 5)
```

In [9]:
```
print("Number of dimensions:")
print(df.ndim)
```

```
Number of dimensions:
2
```

In [10]:
```
print("Column names:")
print(df.columns)
```

```
Column names:
Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
       'petal width (cm)', 'species'],
      dtype='object')
```

In [11]:
```
print(df.describe())
```

```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)
count        150.000000
mean           1.199333
std            0.762238
min            0.100000
25%            0.300000
50%            1.300000
75%            1.800000
max            2.500000
```

In [12]:
```
print("Missing values in each column:")
print(df.isnull().sum())
```

```
Missing values in each column:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
dtype: int64
```

In [13]:
```
print("Total missing values:")
print(df.isnull().sum().sum())
```

```
Total missing values:
0
```

In [14]:
```
print("Number of duplicate rows:")
print(df.duplicated().sum())
```

```
Number of duplicate rows:
1
```

In [17]:
```
print("Create a NumPy array from a column")
petal_length = df["petal length (cm)"].to_numpy()

print("NumPy array:")
print(petal_length)
```

```
Create a NumPy array from a column
NumPy array:
[1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 1.5 1.6 1.4 1.1 1.2 1.5 1.3 1.4
 1.7 1.5 1.7 1.5 1.  1.7 1.9 1.6 1.6 1.5 1.4 1.6 1.6 1.5 1.5 1.4 1.5 1.2
 1.3 1.4 1.3 1.5 1.3 1.3 1.3 1.6 1.9 1.4 1.6 1.4 1.5 1.4 4.7 4.5 4.9 4.
 4.6 4.5 4.7 3.3 4.6 3.9 3.5 4.2 4.  4.7 3.6 4.4 4.5 4.1 4.5 3.9 4.8 4.
 4.9 4.7 4.3 4.4 4.8 5.  4.5 3.5 3.8 3.7 3.9 5.1 4.5 4.5 4.7 4.4 4.1 4.
 4.4 4.6 4.  3.3 4.2 4.2 4.2 4.3 3.  4.1 6.  5.1 5.9 5.6 5.8 6.6 4.5 6.3
 5.8 6.1 5.1 5.3 5.5 5.  5.1 5.3 5.5 6.7 6.9 5.  5.7 4.9 6.7 4.9 5.7 6.
 4.8 4.9 5.6 5.8 6.1 6.4 5.6 5.1 5.6 6.1 5.6 5.5 4.8 5.4 5.6 5.1 5.1 5.9
 5.7 5.2 5.  5.2 5.4 5.1]
```

In [18]:
```
print("Array type:", type(petal_length))
print("Array shape:", petal_length.shape)
print("Array dimensions:", petal_length.ndim)
```

```
Array type: <class 'numpy.ndarray'>
Array shape: (150,)
Array dimensions: 1
```

In [19]:
```
print("Total:", np.sum(petal_length))
print("Minimum:", np.min(petal_length))
print("Maximum:", np.max(petal_length))
print("Average:", np.mean(petal_length))
```

```
Total: 563.7
Minimum: 1.0
Maximum: 6.9
Average: 3.7580000000000005
```

In [21]:
```
print("By using array built in methods")
print("Total:", petal_length.sum())
print("Minimum:", petal_length.min())
print("Maximum:", petal_length.max())
print("Average:", petal_length.mean())
```

```
By using array built in methods
Total: 563.7
Minimum: 1.0
Maximum: 6.9
Average: 3.7580000000000005
```

In [22]:
```
column = df["petal length (cm)"]

print("Count:", column.count())
print("Total:", column.sum())
print("Minimum:", column.min())
print("Maximum:", column.max())
print("Average:", column.mean())
print("Median:", column.median())
print("Standard deviation:", column.std())
print("Variance:", column.var())
print("25th percentile:", column.quantile(0.25))
print("50th percentile:", column.quantile(0.50))
print("75th percentile:", column.quantile(0.75))
print("Unique values:", column.nunique())
print("Mode:", column.mode().tolist())
```

```
Count: 150
Total: 563.7
Minimum: 1.0
Maximum: 6.9
Average: 3.7580000000000005
Median: 4.35
Standard deviation: 1.7652982332594667
Variance: 3.1162778523489942
25th percentile: 1.6
50th percentile: 4.35
75th percentile: 5.1
Unique values: 43
Mode: [1.4, 1.5]
```

In [23]:
```
duplicates = df[df.duplicated()]

print("Duplicate rows:")
print(duplicates)
```

```
Duplicate rows:
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
\
142                5.8               2.7                5.1               1.9

       species
142  virginica
```

In [24]:
```
duplicates = df[df.duplicated(keep=False)]

print("All rows involved in duplication:")
print(duplicates)
```

```
All rows involved in duplication:
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
\
101                5.8               2.7                5.1               1.9
142                5.8               2.7                5.1               1.9

       species
101  virginica
142  virginica
```

In [25]:
```
print("Duplicate row indexes:")
print(df.index[df.duplicated()].tolist())
```

```
Duplicate row indexes:
[142]
```

In [ ]:
```

```